

Infinite-Horizon Policy Evaluation

Reinforcement Learning

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1 Infinite-Horizon Discounted Policy Evaluation (Sutton Notation)

We are taking only infinite horizon, discounted and stationary, Markov random policy, i.e.

$$\pi^\infty = (\pi, \pi, \pi, \dots)$$

1.1 The Environment Model (Primitive Object)

The environment model is specified by a stochastic model

$$p(s', r \mid s, a) = \mathbb{P}(S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a)$$

This defines a joint distribution over next state and reward.

$$(S_{t+1}, R_{t+1}) \mid (S_t = s, A_t = a) \sim p(\cdot, \cdot \mid s, a)$$

This is one conditional random vector.

From this we can recover marginals:

$$p(s' \mid s, a) = \sum_{r \in \mathcal{R}} p(s', r \mid s, a)$$

$$p(r \mid s, a) = \sum_{s' \in \mathcal{S}} p(s', r \mid s, a)$$

1.2 Expected Reward Function

$$r(s, a) = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] = \sum_{r \in \mathcal{R}} r p(r \mid s, a) = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p(s', r \mid s, a)$$

This is the expected reward we are going to get given that we are starting from state $s \in \mathcal{S}$ and taking action $a \in \mathcal{A}(s)$.

This is a deterministic function, a function of current state and the action we are going to take.

In general in Puterman book we directly assume we know this function. There is no randomness in obtaining reward when we are choosing action a from state s .

But here in Sutton book and in general RL we assume reward is a random variable, i.e., random each time when we are taking action a from state s , because we are accounting for noise in observation or we do not know the fully reward model.

1.3 Returns

If the sequence of rewards received after time step t is denoted by

$$R_{t+1}, R_{t+2}, R_{t+3}, \dots,$$

then we define a random variable G_t which is a specific function of the reward sequence.

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

Here γ is the discount parameter.

Here G_t is a random variable, since each reward R_{t+k+1} is random. We do not know their specific values, and therefore the summation of these rewards is also random.

1.4 Value Function and Action-Value Function for Policy π

The value function for a given policy π is a fixed deterministic vector, essentially a value for each state. It is the expected reward we are going to receive starting from state s and following policy π .

$$V^\pi(s) = \mathbb{E}^\pi[G_t \mid S_t = s] = \mathbb{E}^\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s \right].$$

Similarly, we define Action-Value function which is expected return starting from state s , taking action a , and thereafter following policy π :

$$Q^\pi(s, a) = \mathbb{E}^\pi[G_t \mid S_t = s, A_t = a] = \mathbb{E}^\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a \right].$$

1.5 Deriving the Value Function

Let us solve the Value function first.

$$\begin{aligned} V^\pi(s) &= \mathbb{E}^\pi[G_t \mid S_t = s] \\ &= \mathbb{E}^\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \mathbb{E}^\pi[R_{t+1} \mid S_t = s] + \gamma \mathbb{E}^\pi[G_{t+1} \mid S_t = s] \end{aligned}$$

We will solve these individually.

First Term

$$\mathbb{E}^\pi[R_{t+1} \mid S_t = s]$$

This is the expected immediate reward we are going to receive starting from state s . We can choose action a according to the policy, and depending on the action we receive the reward.

$$\mathbb{E}^\pi[R_{t+1} \mid S_t = s] = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \mathbb{E}^\pi[R_{t+1} \mid S_t = s, A_t = a]$$

We just applied law of total expectation here.

$$\begin{aligned} &= \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{r \in \mathcal{R}} r p(r \mid s, a) \\ &= \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p(s', r \mid s, a) \\ &= \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) r(s, a) \end{aligned}$$

We can think this as repeatedly conditioning and getting the value. Given that we are starting from s , we can take any action, so we multiply by $\pi(a \mid s)$. Now given that we are at state s and taking action a , from the joint probability we can get any rewards $R_{t+1} = r$ and we could go to any states $S_{t+1} = s'$. We have taken expectation over all this randomness and we reached:

$$\mathbb{E}^\pi[R_{t+1} \mid S_t = s] = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) r(s, a)$$

Second Term

$$\gamma \mathbb{E}^\pi[G_{t+1} \mid S_t = s]$$

This is essentially: given that we are in state s currently, what would be the expected future discounted reward we could get.

Again we break down into multiple things.

First, we could take any action a from $\mathcal{A}(s)$:

$$\gamma \mathbb{E}^\pi[G_{t+1} \mid S_t = s] = \gamma \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \mathbb{E}^\pi[G_{t+1} \mid S_t = s, A_t = a]$$

Now we are at state s and taken action a , we could randomly get into state s' from there.

$$= \gamma \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{s' \in \mathcal{S}} p(s' \mid s, a) \mathbb{E}^\pi[G_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s']$$

Now since we are at state s' , due to Markov property, it does not matter whether we started from $S_t = s$ or took action $A_t = a$. If we know we are at state s' at time $t + 1$, then all the history of how we arrived there does not matter for future reward.

$$= \gamma \sum_{a \in \mathcal{A}(s)} \pi(a | s) \sum_{s' \in \mathcal{S}} p(s' | s, a) \mathbb{E}^\pi[G_{t+1} | S_{t+1} = s']$$

But

$$\mathbb{E}^\pi[G_{t+1} | S_{t+1} = s'] = V^\pi(s')$$

Thus,

$$= \gamma \sum_{a \in \mathcal{A}(s)} \pi(a | s) \sum_{s' \in \mathcal{S}} p(s' | s, a) V^\pi(s')$$

Expanding transition via joint model:

$$= \gamma \sum_{a \in \mathcal{A}(s)} \pi(a | s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r | s, a) V^\pi(s')$$

Final Value Function (Multiple Versions)

Putting everything together:

$$V^\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a | s) \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) V^\pi(s') \right]$$

Another version:

$$= \sum_{a \in \mathcal{A}(s)} \pi(a | s) \left[\sum_{r \in \mathcal{R}} r p(r | s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) V^\pi(s') \right]$$

Fully expanded joint version:

$$= \sum_{a \in \mathcal{A}(s)} \pi(a | s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r | s, a) [r + \gamma V^\pi(s')]$$

1.6 Action-Value Function

This is nothing but what expected total discounted rewards we are going to receive when we are starting from state s and taking action a .

$$Q^\pi(s, a) = \mathbb{E}^\pi[G_t | S_t = s, A_t = a]$$

We know

$$\begin{aligned} V^\pi(s) &= \mathbb{E}^\pi[G_t \mid S_t = s] = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \mathbb{E}^\pi[G_t \mid S_t = s, A_t = a] \\ &= \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) Q^\pi(s, a) \end{aligned}$$

From the above value function versions we can write

$$Q^\pi(s, a) = r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, a) V^\pi(s')$$

This is essentially: the immediate expected reward we are going to get starting from s and taking action a , which is $r(s, a)$, and then moving forward to the next state s' (which is random), and then the expected total discounted reward we get starting from s' .